

Week 5 Assignment

Due: Before the Week 6 class on Thursday, Oct. 1

Data: gender_voice_data.csv

Target: gender

ML Objective: Identify the gender of a speaker based upon the statistics from their recording.

Submit: Your solution *wk5_solution.py*, the output from your code *wk5_output.txt*, and the summary review *wk5_opinion_review.pdf*.

Data were extracted from 3,168 individuals who submitted voice recordings. Each recording was summarized using 20 audio metrics. The dataset includes the speaker's gender and these 20 metrics.

This exercise is to build an ML model (logistic regression) to automatically identify the speaker's gender based on their audio metrics. Please develop your solution following the approach used in the Week 5 class exercise. Use Cursor following the checklist "*Cursor_Project_Checklist*". Also set your Cursor Agent Model to "*Composer 2.5*".

Create your project folder using the zip file distributed for the class exercise:

"*Wk5_Class_Exercise.zip*." To keep the exercise and assignment separate, rename the project folder to "*Wk5_Assignment*".

This assignment is a larger version of the class exercise. Both use a binary target, but the assignment has 20 predictors instead of 4. Follow the steps in the class exercise, and save your solution output to a text file "*wk5_output.txt*".

After you are satisfied with your solution, ask a *new* Cursor Agent to review your Python code and its output. Have the agent summarize the results, state what worked well, suggest improvements, and save the review as "*wk5_opinion_review.pdf*".